

Pretrained Encoder + JacobianODE – Report

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Date: 2026-03-07

W&B project: JacobianODE/Lorenz_ETD_JacODE

Best run ID: k6relrt4 ($\lambda=1e-06$)

DATA

System: Lorenz

dt = 0.0150

Observed indices: [0]

N_delays = 5, delay_spacing = 10

n_obs (delay-embedded) = 5

Obs noise = 0.01

Num ICs = 32, N_periods = 12

ENCODER

Type: EigentimeDelayAdapter

n_latent = 3

Frozen: frozen

Encoder params: 0

JACOBIAN ODE

prediction_steps = 30

traj_init_steps = 15

Jac MLP params: 6,578,185

Total params: 6,578,185

MODEL SELECTION

Sweep lambdas: [0, 1e-06, 1e-05, 0.0001, 0.001, 0.01, 0.1, 1, 10, 100]

Criteria applied: ['C3']

Surviving: 10 / 10

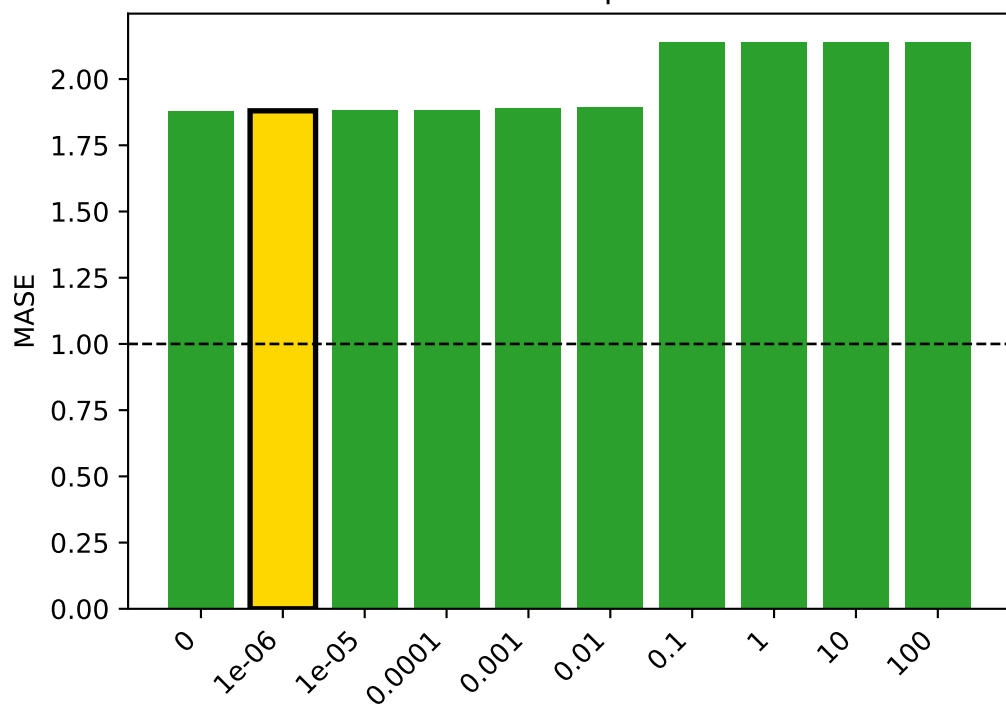
Best traj val loss: 0.018003

Best MASE: 1.8801

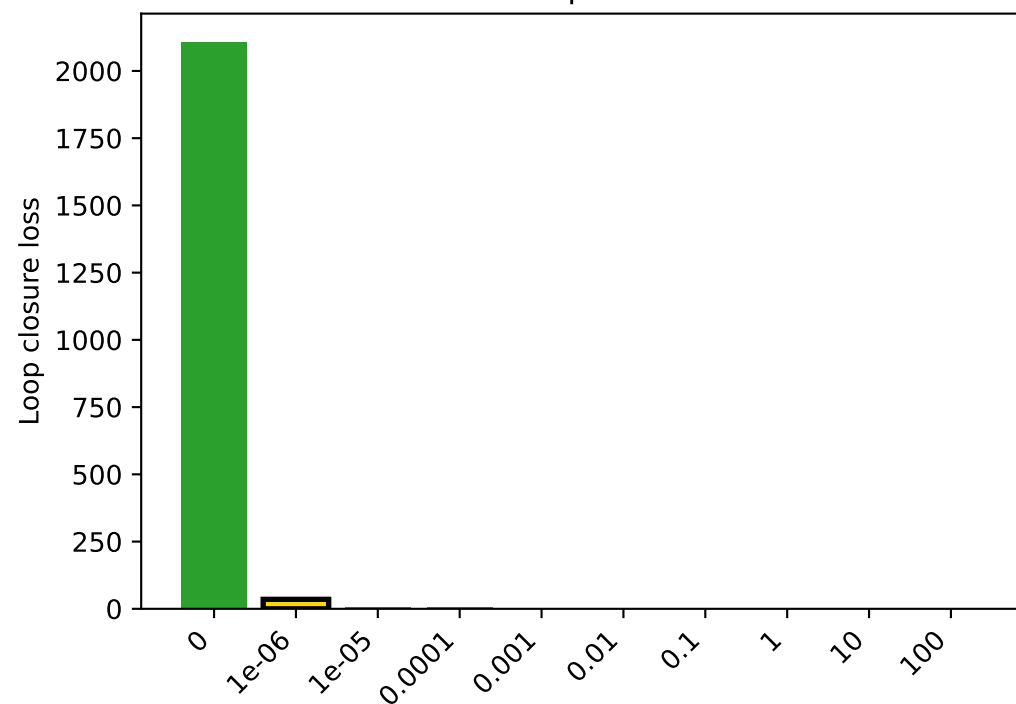
Best loop closure: 35.2264

Model Selection (best lambda=1e-06)

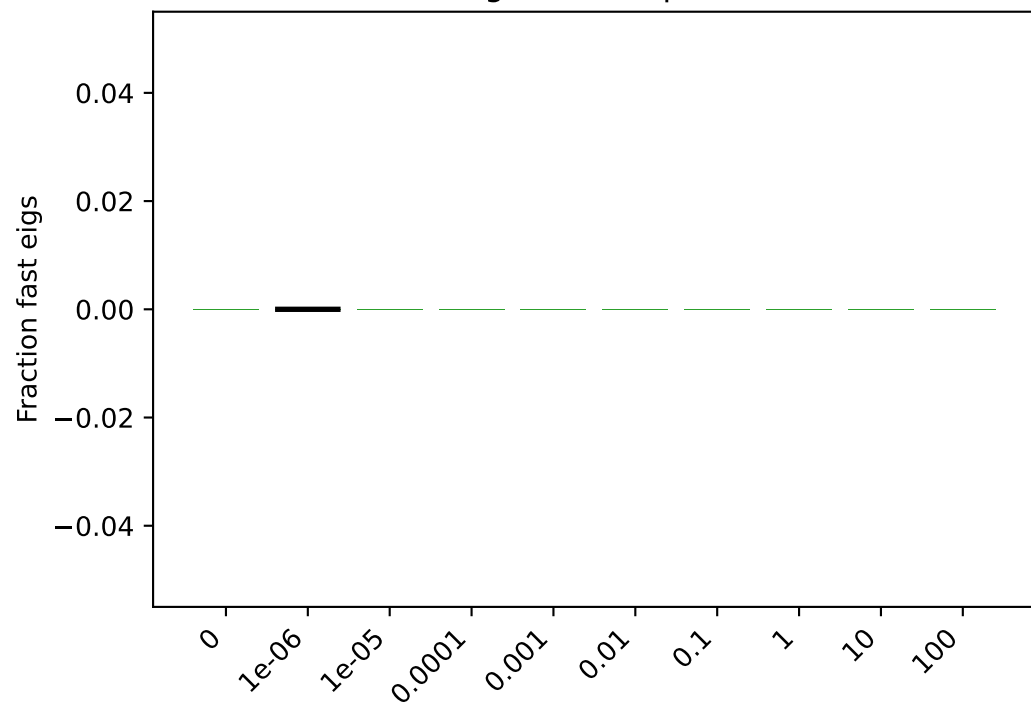
C1: One-step MASE



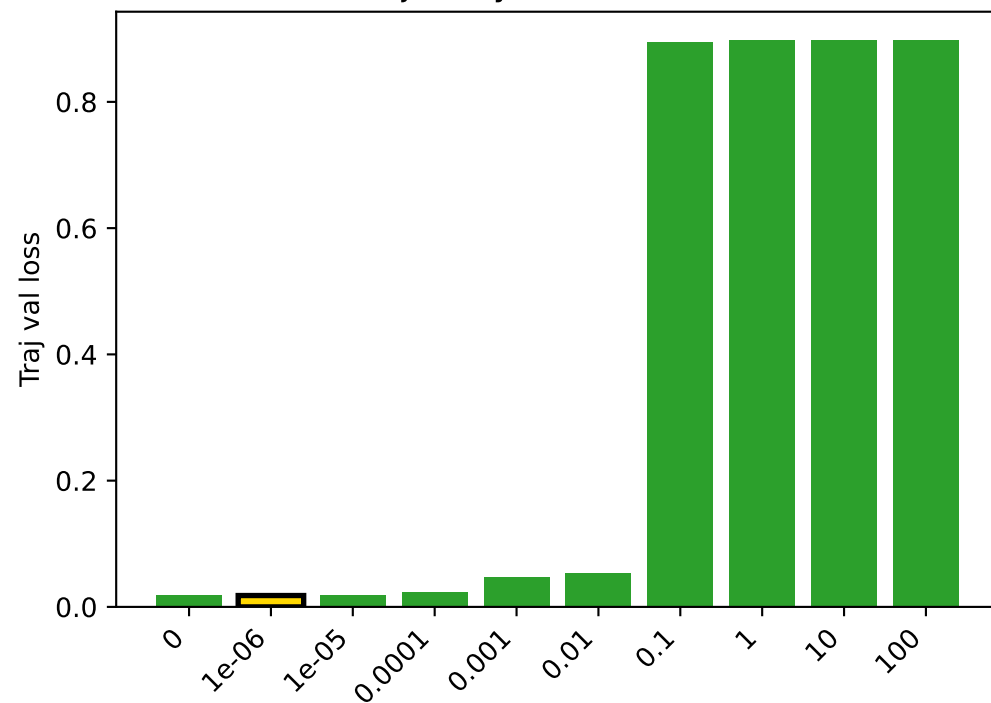
C2: Loop Closure



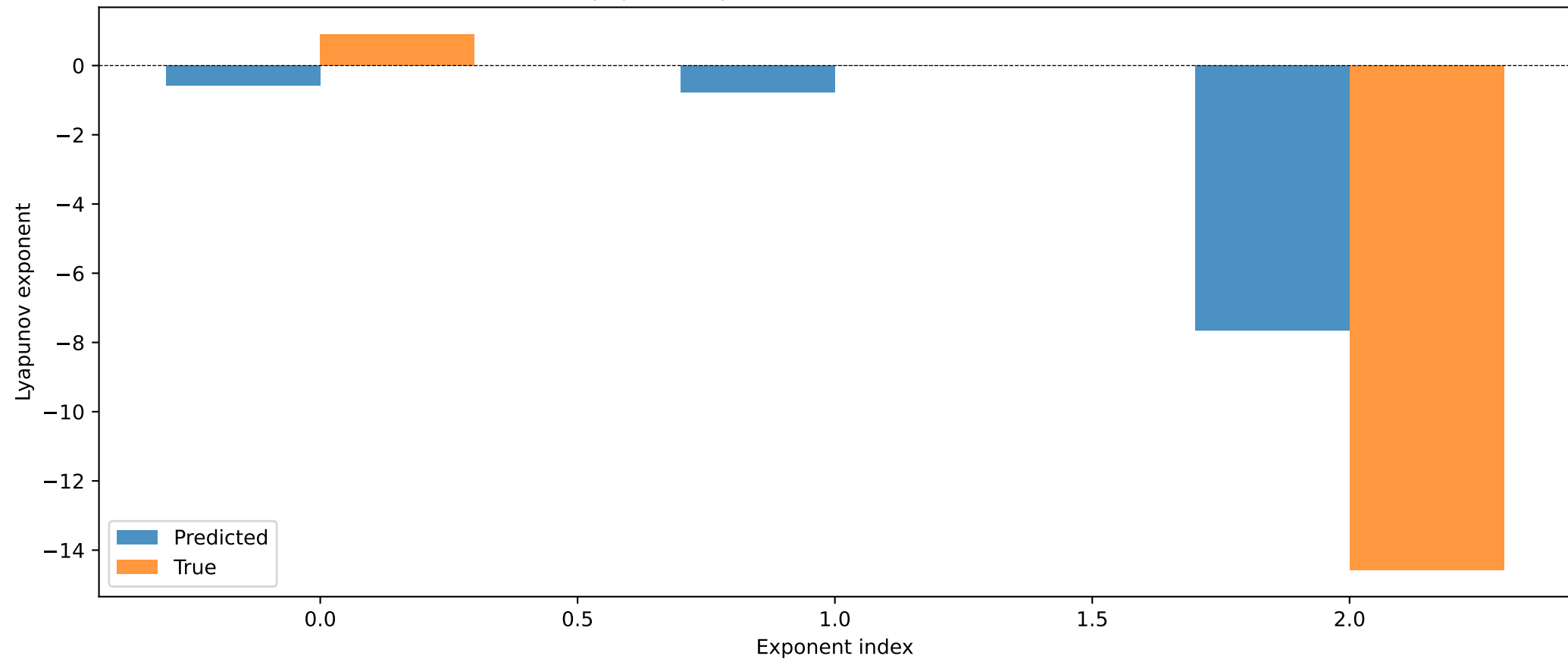
C3: Eigenvalue Spectrum



Trajectory Validation Loss



Lyapunov Spectrum (best lambda=1e-06)



Predicted Lyapunov exponents (best model):

lambda_1 = -0.5736

lambda_2 = -0.7726

lambda_3 = -7.6661

True Lorenz:

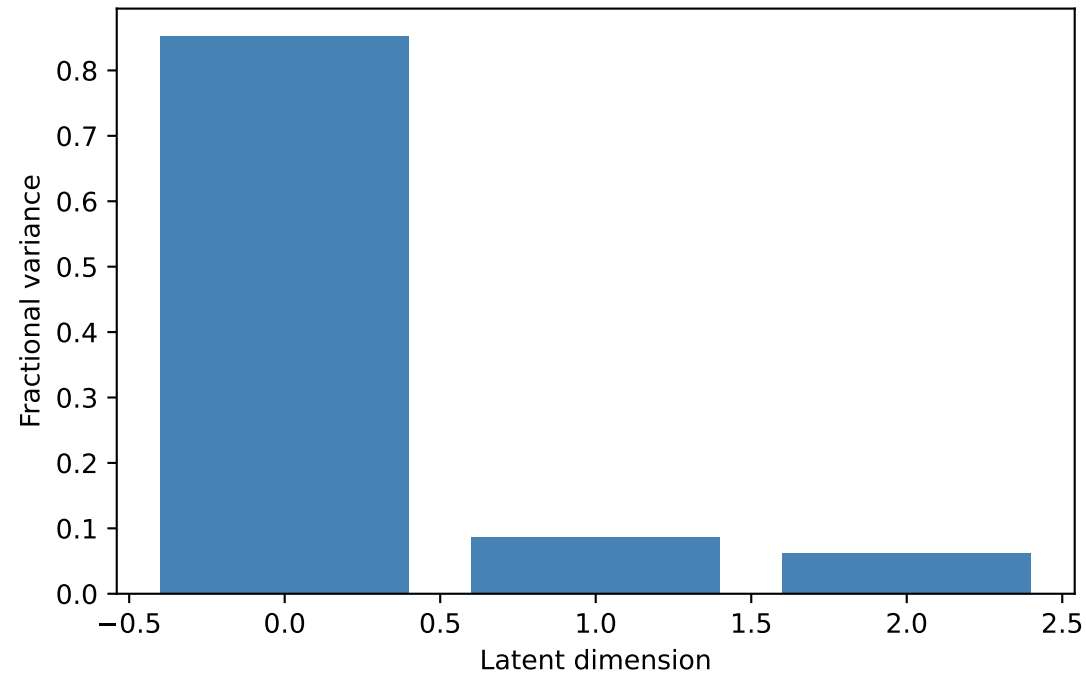
lambda_1 = +0.9100

lambda_2 = +0.0000

lambda_3 = -14.5700

MSE (top 3 exponents): 16.8207

Latent dimension utilisation

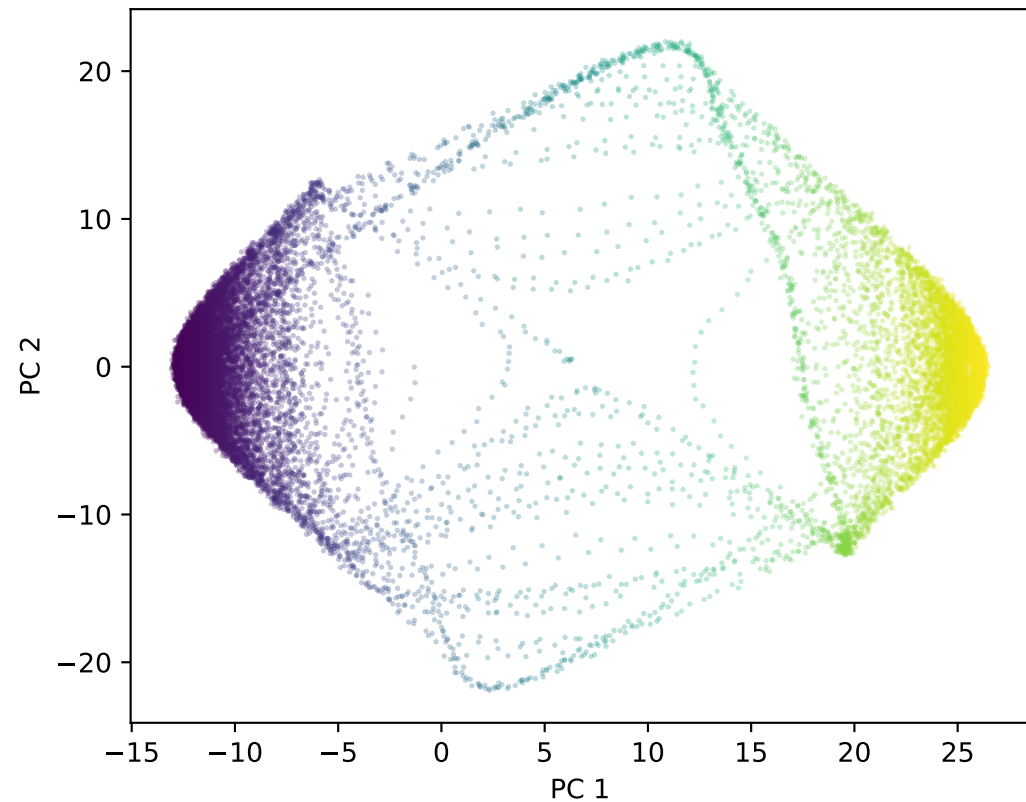


Summary

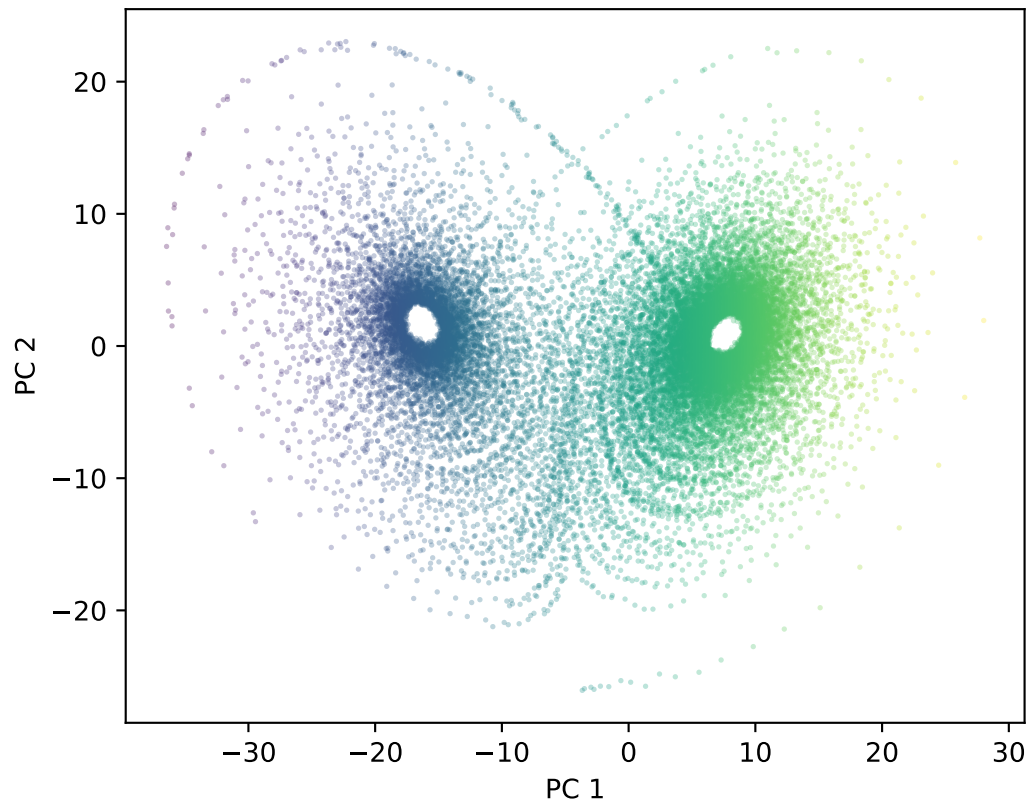
Utilization: 0.473
Participation ratio: 1.356
True state PR: 1.499

PC 1 x PC 2

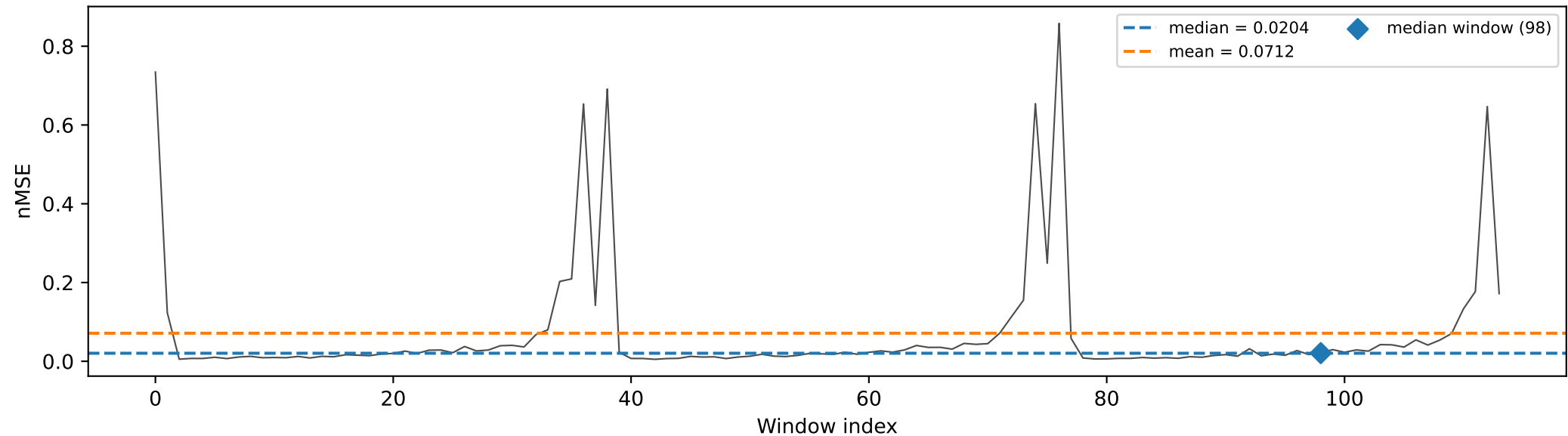
Latents (PR=1.356)



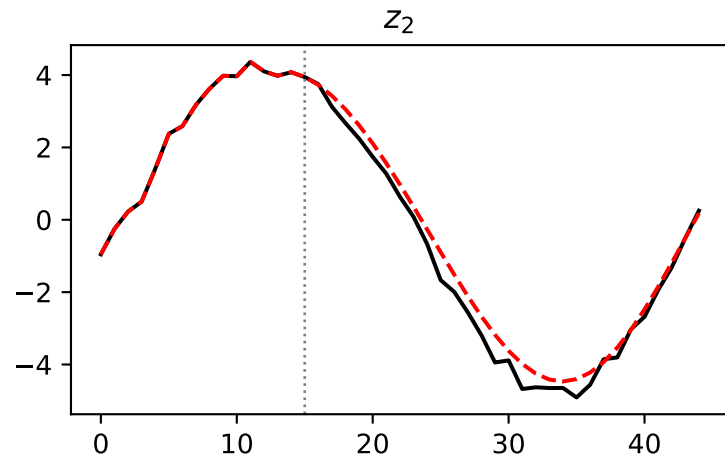
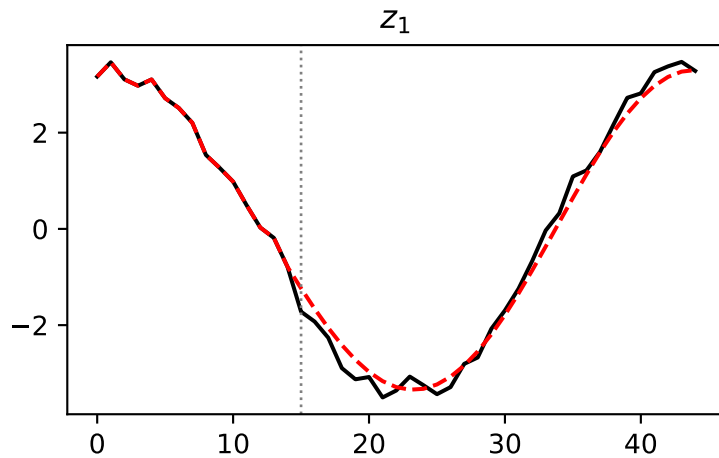
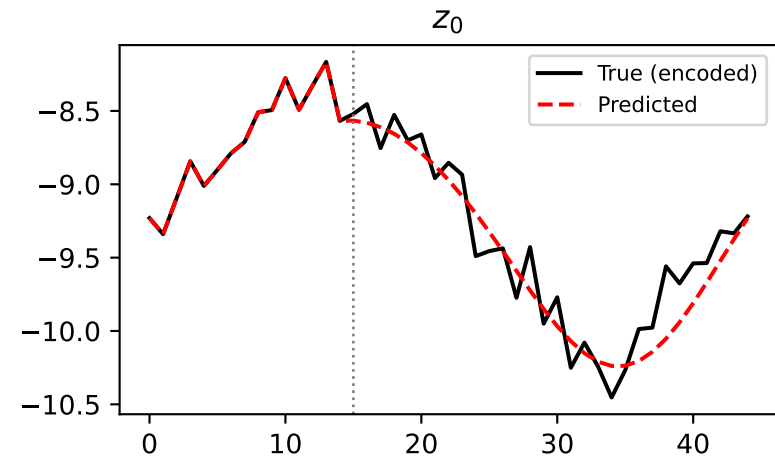
True state (PR=1.499)



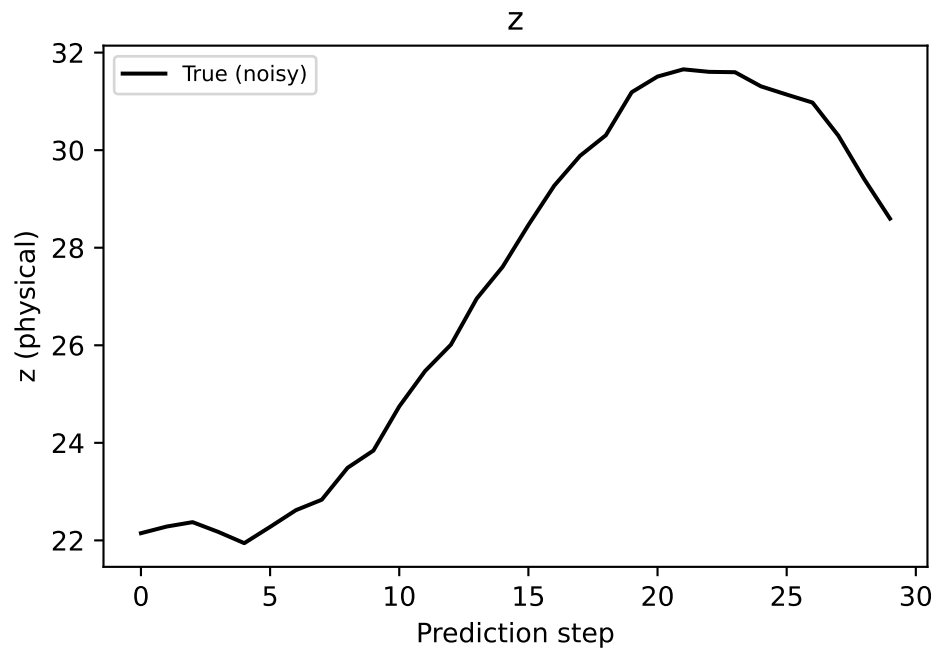
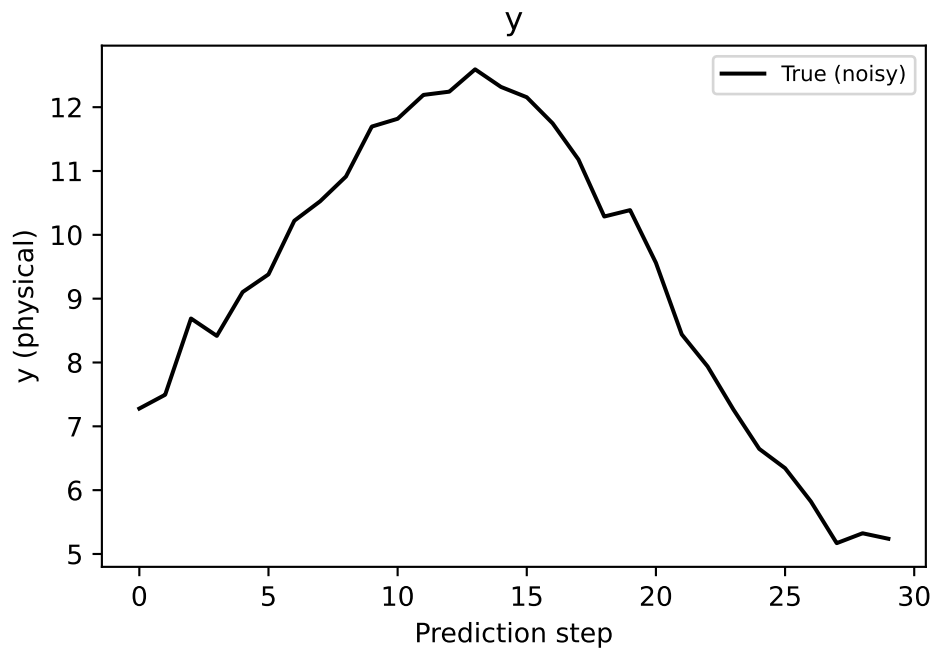
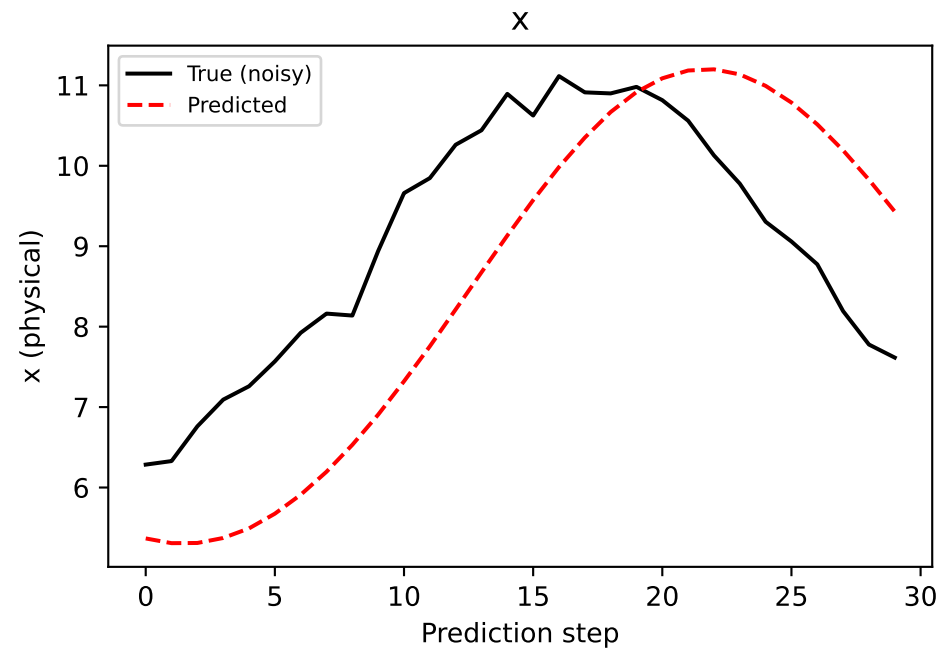
Per-window nMSE (114 windows)



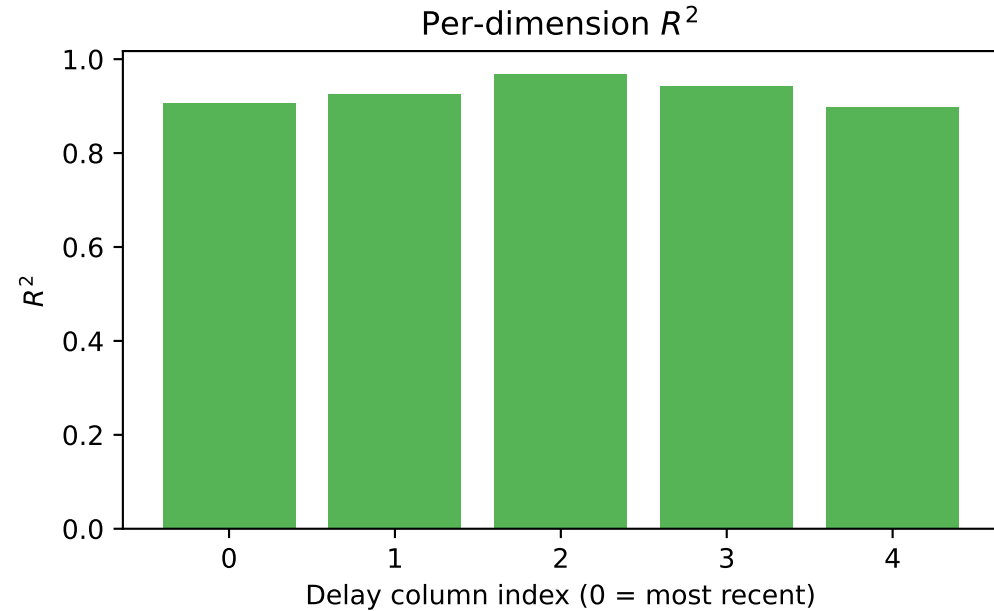
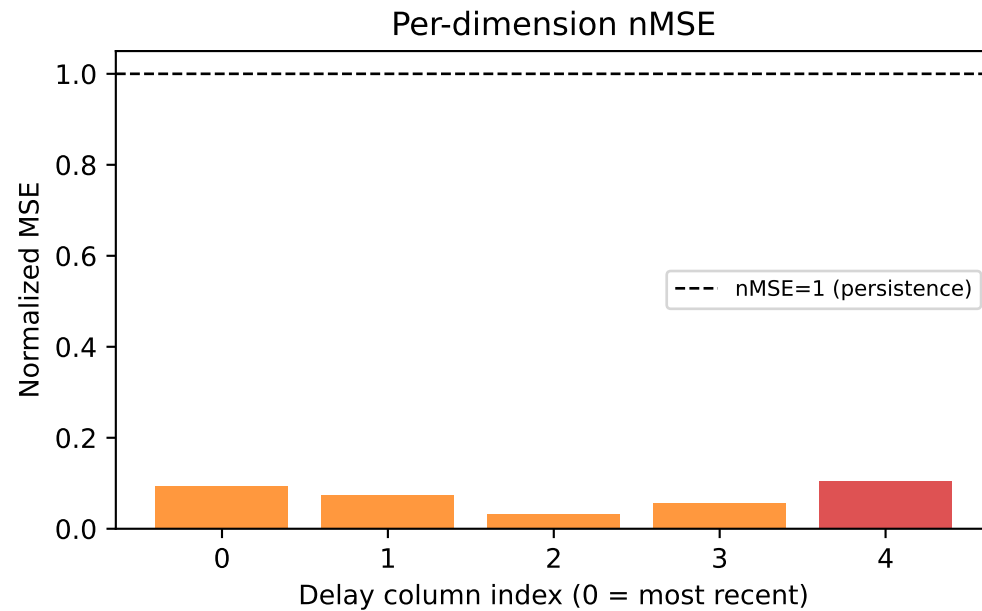
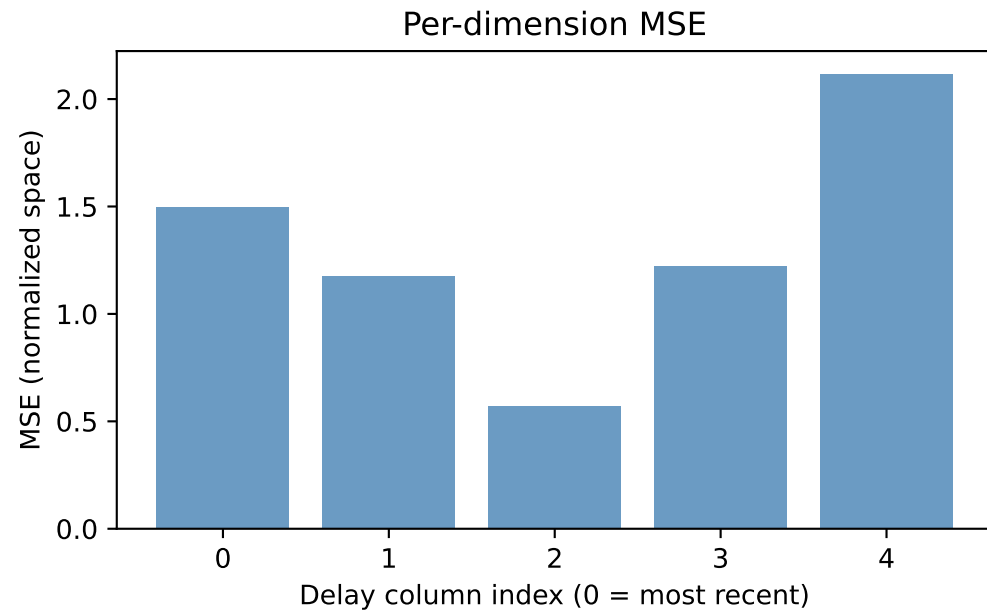
Latent Space: Median-loss window (idx=98, nMSE=0.0207)



Physical Space: Median-loss window (idx=98, nMSE=0.0207)



Prediction error by delay-embedding dimension
(averaged over 114 windows, 3 test trajectories)



Per-dimension prediction error table

Dim	Lag (dt)	MSE	nMSE	R ²	Var(true)
0	0.0000	1.498680	0.093152	0.9068	16.088516
1	0.1502	1.173035	0.074017	0.9260	15.848193
2	0.3004	0.572266	0.031203	0.9688	18.340275
3	0.4506	1.222645	0.056716	0.9433	21.557226
4	0.6008	2.117553	0.103422	0.8966	20.474859

Overall nMSE (mean across dims): 0.071702
Dim 0 (current x) nMSE: 0.093152
Dim 4 (oldest delay) nMSE: 0.103422